15.Implement Iris Flower Classification using Naive Bayes classifier

CODING:

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

iris = load_iris()

X = iris.data

y = iris.target

print("Iris Dataset")

print("------------")

print("Features:", iris.feature_names)

print("Classes :", iris.target_names)

print("Total Samples:", len(X))

X_train, X_test, y_train, y_test = train_test_split( X, y,

    test_size=0.3,

    random_state=42,

    stratify=y)

model = GaussianNB()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("\nNaive Bayes Classification Results")

print("----------------------------------")

print("Actual Values   :", y_test)

print("Predicted Values:", y_pred)

print("Accuracy       :", accuracy)

print("\nConfusion Matrix:")
```

```
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")

print(classification_report(  y_test,

   y_pred,

   target_names=iris.target_names))

new_flower = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(new_flower)

print("\nNew Flower Prediction:")

print("Predicted Species:", iris.target_names[prediction[0]])
```

OUTPUT:

```
Iris Dataset
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Features: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']
Classes : ['setosa' 'versicolor' 'virginica']
Total Samples: 150

Naive Bayes Classification Results
----------------------------------
Actual Values   : [2 1 2 1 2 2 1 1 0 2 0 0 2 2 0 2 1 0 0 0 1 0 1 2 2 1 1 1 1 0 2 2 1 0 2 0 0
 0 0 1 1 0 2 2 1]
Predicted Values: [2 1 1 1 2 2 1 1 0 2 0 0 2 2 0 2 1 0 0 0 1 0 1 2 2 1 1 1 1 0 1 2 1 0 2 0 0
 0 0 2 1 0 1 2 1]
Accuracy        : 0.9111111111111111

Confusion Matrix:
[[15  0  0]
 [ 0 14  1]
 [ 0  3 12]]

Classification Report:
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        15
  versicolor       0.82      0.93      0.88        15
   virginica       0.92      0.80      0.86        15

    accuracy                           0.91        45
   macro avg       0.92      0.91      0.91        45
weighted avg       0.92      0.91      0.91        45


New Flower Prediction:
Predicted Species: setosa
```